# BATTERY-DISCHARGE INDICATOR

## IC4484B915 FOR 24-, 36- AND 48-VOLT OPERATION

*Before any adjustments, servicing, parts replacement or any other act is performed requiring physical contact with the electrical working components or wiring of this equipment, DISCONNECT THE BATTERY AND DISCHARGE CAPACITOR 1C.*

## FUNCTION

The purpose of the battery-discharge indicator is to monitor the open-circuit voltage of a lead-acid battery and to provide the following functions:

1. Indicate the approximate state of charge. (Optional)
2. Switch a warning light on and off to indicate a low battery.
3. Turn the warning light on continuously and disable the lift-pump contactor coil to indicate a discharged battery.
4. Disable the forward contactor coil, if the operator discharges the battery still further. (Optional)

Since checking the unit requires measurement accuracy beyond that of meters generally available in the field, the unit is preset at the factory. Input terminals are provided for battery voltages most often used in the Industrial Truck industry.

The relation between open-circuit volts, state of charge, and specific gravity used on the battery-discharge indicator is shown on the graph in Fig. 1. A small variation between specific gravity and open-circuit volts-per-cell does occur. A specific gravity of 1.150 is the generally recommended low battery condition for optimum battery life. Refer to your battery representative for specific recommendations for your batteries.

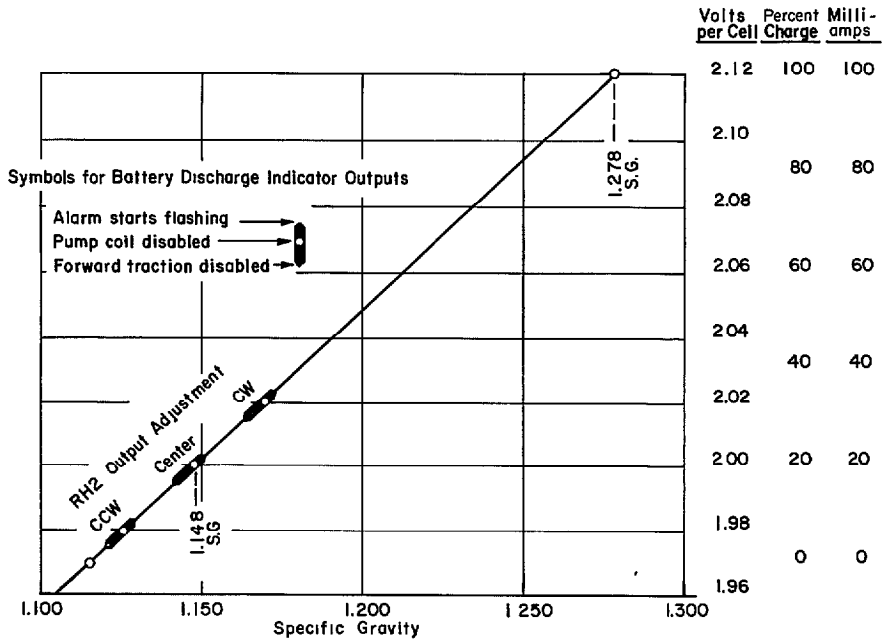



*Fig. 1. Specific gravity/volts per cell and percent charge*

The information contained herein is intended to assist truck users and dealers in the servicing of SCR control furnished by the General Electric Company. It does not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance.

Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purpose, the matter should be referred to the truck manufacturer through his normal service channels, not directly to General Electric Company.

## OPERATION

The battery terminal voltage is monitored using a peak-reading circuit that is filtered with a time constant (0.3 seconds) much shorter than the time constant for open-circuit battery voltage to recover (two minutes). The peak-reading circuit will decay at a rate equivalent to discharging a fully charged battery in three hours. Each time the load is removed, the peak-reading circuit will be reset upwards to the actual open-circuit voltage. The peak-reading circuit output is used to provide a low-current output which is proportional to the charge remaining on the battery and may be used for a "fuel gauge" meter. Fractions of the peak-reading circuit output are also compared to a voltage, adjustable by RH2, to provide alarm, discharge and disable outputs.

Terminals 6 and 5 can be used for a 1-milliampere full-scale meter which will show 100-percent charge at full scale and empty (80-percent discharge) at 0.2 milliamperes.

When the peak-reading circuit falls below the alarm point, output terminal 4 switches the light on and off at a rate of approximately 3 Hertz. As the peak-reading circuit continues to drop, the switching rate increases until the discharge point is reached. At this time, output terminal 4 switches on continuously and the light stays on to signal that the battery has reached its recommended discharge point. The light will also come on continuously if the operator calls for heavy loads, such as SCR stall. This warns the operator that battery power is being consumed at a higher than normal rate. At the discharge point, the pump contactor can be disabled either with output terminal 3 or with output terminals 11 and 10 if a pump time delay is used.

If the peak-reading circuit continues to drop, the disable output terminal 2 will turn off. This may be used with a coil driver to disable the forward traction contactor.

If the battery volts (peak-reading circuit) recovers a small amount above the disable point, output terminal 3 will be turned on and output terminal 4 will be turned off, allowing additional full truck operation.

The use of the peak-reading circuit introduces three limitations which will not be a problem in normal Industrial Truck applications:

1. The battery must be disconnected from the control whenever the battery is to be connected to a charger. If a battery is only partially charged and immediately connected to the control, the peak-reading circuit will start higher than normal.
2. If the average load current exceeds the three-hour discharge rate, the battery will discharge faster than the peak-reading circuit will decay. The battery will become discharged before the battery-discharge-indicator outputs so indicate.
3. If a load less than the three-hour discharge rate remains on continuously for a long period of time, the battery-discharge-indicator outputs will show a discharge condition prematurely but will reset when the load is removed.

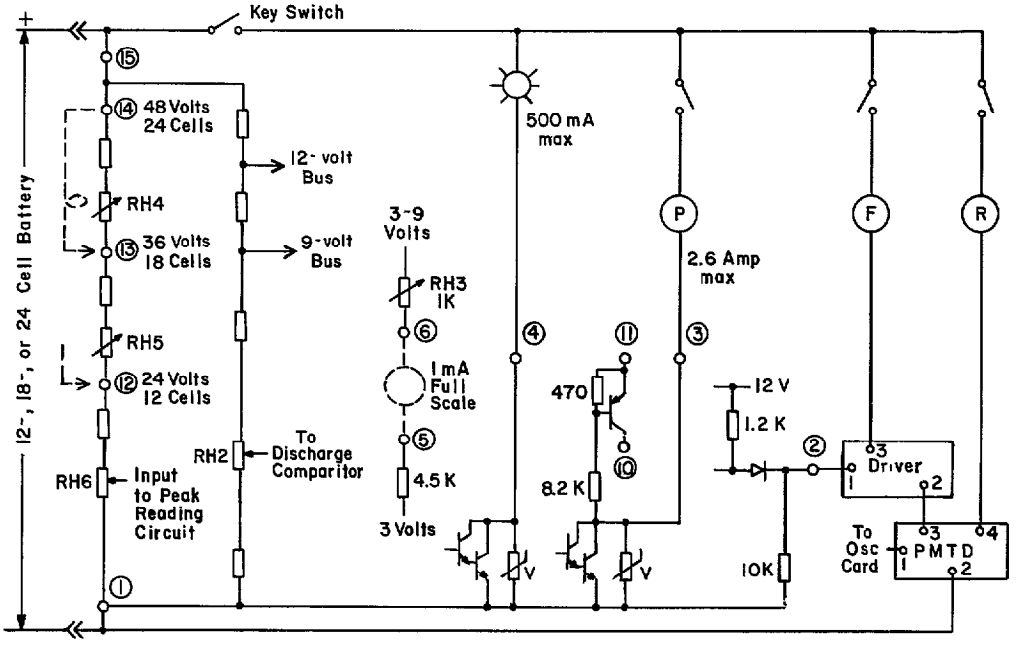



*Fig. 2. Connection diagram – battery discharge indicator*

## CONNECTIONS

Connect the battery-discharge indicator per Fig. 2. Add a jumper from terminal 14 to 13 or 12 according to the number of cells.

If the pump coil is operated by a pump time delay, connect the pump coil per Fig. 3.

## ADJUSTMENTS

Trimpots RH4, RH5 and RH6 have been set so that the volts-per-cell corresponding to zero charge gives zero output at terminal 6 to 5. Trimpots have been sealed at the factory. DO NOT ALTER.

RH2 is a single-turn pot with ample adjustment range to alter the peak-reading-circuit voltage that operates the battery-discharge-indicator outputs. Nominally, this trimpot should be set at the midpoint. The alarm and disable points are fixed incremental voltages above and below this point.

RH3 has been set and sealed at the factory to give 1 milliampere at the volts-per-cell corresponding to a full charge. This can be changed without affecting the calibration of the other battery-discharge-indicator outputs.

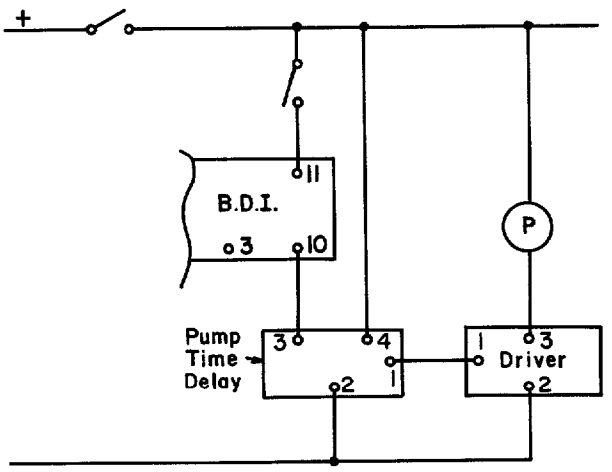



*Fig. 3. Alternative connection to pump when pump time delay is used*

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